

YUXUAN (VINCENT) ZHANG

PH H1 477, Lausanne, Switzerland

yuxuan.zhang@epfl.ch ♦ [personal page](#) ♦ [google scholar](#)

RESEARCH INTERESTS

Theme: *toward verifiable and practical quantum advantage*

Keywords: quantum computation · quantum many-body physics · artificial intelligence · algorithm development & verification · quantum simulation

EDUCATION

Ph.D. in Physics, The University of Texas at Austin

Aug. 2017 – Aug. 2023

Advisors: Scott Aaronson and Andrew C. Potter

Thesis: *Exploring quantum matter in the era of quantum computers*

B.S. in Physics, University of California, Santa Barbara

Sep. 2013 – Jun. 2016

Graduated with Highest Honors (top 2.5% of the class)

EMPLOYMENT

Assistant Professor (incoming), National University of Singapore

from Dec. 2026

Joint appointment: *Department of Computer Science* (primary) and *Department of Physics*

Postdoctoral Fellow, EPFL & Princeton

Dec. 2025 – present

Supported by an [ERC Consolidator Grant](#)

Host: Dmitry Abanin

Research Scientist, BlueQubit Inc.

Dec. 2025 – present

Role: quantum algorithm development (part-time)

Postdoctoral Fellow, U of T & [Vector Institute for Artificial Intelligence](#)

Sep. 2023 – Dec. 2025

Supported by the [CQIQC Fellowship](#) (Sep. 2023 – Aug. 2025) and the [Quantum Software Consortium Grant](#) (Sep. 2025 – Dec. 2025)

Hosts: Yong-Baek Kim, Juan Felipe Carrasquilla Álvarez, and Dvira Segal

Focus: synergy between **quantum information**, **quantum matter**, and **artificial intelligence**

SELECTED PUBLICATIONS

Observation of a non-Hermitian supersonic mode on a trapped-ion quantum processor

[Nat Comm](#)

Y. Zhang (first author), J. Carrasquilla, and Y. B. Kim; *Nature Communications* 16, 3286 (2025)

Classical simulability of circuits with shallow magic depth

[PRX-Q](#)

Y. F. Zhang and **Y. X. Zhang** (corresponding author); *PRX Quantum* 6, 010337

Probing mixed-state phases on a quantum computer via Rényi correlators and variational decoding

[Nat Comm](#)

Y. Zhang (first author), T. Hsieh, Y. B. Kim, and Y. Zou; *Nature Communications* (2026)

Scalable quantum dynamics compilation via quantum machine learning

[PRR](#)

Y. Zhang (first author), R. Wiersema, Y. B. Kim, J. Carrasquilla, and L. Cincio; *Phys. Rev. Research* 8, 023128

Quantum volume for photonic quantum processors

[PRL](#)

Y. Zhang (first author), D. Niu, A. Shabani, and H. Shapourian; *Physical Review Letters* 130, 110602

QED-driven QAOA for network-flow optimization

[Quantum](#)

Y. Zhang (first author), R. Zhang, and A. C. Potter; *Quantum* 5, 510 (2021)

PUBLICATIONS

- Evaluating LLMs in Scientific Discovery [arXiv](#)
Z. Song et al., including **Y. Zhang**
Featured by [Science news](#)
Under review, *Nat Comput Sci*.
- Heuristic Quantum Advantage with Peaked Circuits [arXiv](#)
H. Gharibyan, M. Z. Mullath, N. E. Sherman, V. P. Su, H. Tepanyan, and **Y. Zhang**
- On verifiable quantum advantage with peaked circuit sampling [arXiv](#)
S. Aaronson, **Y. Zhang** (corresponding author)
Under review, *Physical Review X*
- Complexity and hardness of random peaked circuits [arXiv](#)
Y. Zhang
- Circuit compression for 2D quantum dynamics [arXiv](#)
M. D’Anna, **Y. Zhang** (corresponding author), R. Wiersema, M. Rudolph, and J. Carrasquilla
Under review, *Science Advances*
- Probing mixed-state phases on a quantum computer via Rényi correlators and variational decoding [Nat Comm](#)
Y. Zhang, T. Hsieh, Y.B. Kim, and Y. Zou
Nature Communications (2026)
- Scalable quantum dynamics compilation via quantum machine learning [PRR](#)
Y. Zhang, R. Wiersema, Y. B. Kim, J. Carrasquilla, and L. Cincio
Phys. Rev. Research 8, 023128
- Biorthogonal Neural Network Approach to Two-Dimensional Non-Hermitian Systems [PRL](#)
M. Solinas, B. Barton, **Y. Zhang**, J. Nys, and J. Carrasquilla
Physical Review Letters, 136, 126501
- Mixed-state phase transitions in spin-Holstein models [PRB](#)
B. Min, **Y. Zhang** (co-first author), Y. Guo, D. Segal, and Y. Ashida
Physical Review B, 111, 115123
- Observation of a non-Hermitian supersonic mode on a trapped-ion quantum processor [Nat Comm](#)
Y. Zhang, J. Carrasquilla, and Y. B. Kim
Nature Communications, 16, 3286 (2025)
Featured in [Nature’s Quantum Many-Body Dynamics Collection](#)
- Classical Simulability of Circuits with Shallow Magic Depth [PRX-Q](#)
Y.F. Zhang, and **Y.X. Zhang** (corresponding author)
PRX Quantum, 6, 010337
Featured in the [PRX International Year of Quantum Collection](#)
- Possibility of entanglement of purification to be less than half of the reflected entropy [PRA](#)
J. Couch, P. Nguyen, S. Racz, G. Stratis, **Y. Zhang** (corresponding author)
Physical Review A, 109, 022426
- Sequential quantum simulation of spin chains with a single circuit QED device [PRA](#)
Y. Zhang, S. Shabani, A. Riswadkar, S. Shankar, and A. C. Potter
Physical Review A, 109, 022606
- All-photonic one-way quantum repeaters [npj-QI](#)
D. Niu, **Y. Zhang**, A. Shabani, and H. Shapourian
npj Quantum Information 9, 106

Holographic Quantum Simulation of Entanglement Renormalization Circuits S. Anand, J. Hauschild, Y. Zhang , A. C. Potter, and M. P. Zaletel <i>PRX Quantum</i> 4, 030334	PRX-Q
Quantum volume for photonic quantum processors Y. Zhang , D. Niu, A. Shabani, and H. Shapourian <i>Physical Review Letters</i> , 130, 110602 CLEO conference 2023	PRL
Qubit-efficient simulation of thermal states with quantum tensor networks Y. Zhang , S. Jahanbani, D. Niu, R. Haghshenas, and A. C. Potter <i>Physical Review B</i> , 106, 165126	PRB
Holographic simulation of correlated electrons on a trapped-ion quantum processor D. Niu, R. Haghshenas, Y. Zhang , M. Foss-Feig, G. K. Chan, and A. C. Potter <i>PRX Quantum</i> , 3, 030317	PRX-Q
Straddling-gates problem in multipartite quantum systems Y. Zhang <i>Physical Review A</i> , 105, 062430	PRA
QED driven QAOA for network-flow optimization Y. Zhang , R. Zhang, and A. C. Potter <i>Quantum</i> , 5, 510 (2021)	Quantum
CEPC Conceptual Design Report Volume II: Physics & Detector M. Abbrescia et al., including Y. Zhang	arXiv
CEPC Conceptual Design Report: Volume 1 – Accelerator M. Abbrescia et al., including Y. Zhang	arXiv

PROFESSIONAL EXPERIENCE

Lecturer: School for Near Term Quantum Computing	Benasque, Spain	Mar. 2026
Visitor	Perimeter Institute for Theoretical Physics	Aug. 2023; Mar., May, Jul. 2024; May, Oct. 2025
Program Participant	Kavli Institute for Theoretical Physics	Apr. 2024, Nov. 2025
Academic guest	ETH Zurich	Nov. 2024 – Dec. 2024; Dec. 2025
Visitor	Tsinghua University	Dec. 2023, Dec. 2024
Participant: Condensed Matter Summer School	University of Minnesota	Jun. 2023
Graduate Research Assistant	UT Austin	Jan. 2020 – Aug. 2023
Mentor: The Quantum Collective	UT Austin	Sep. 2022 – May 2023
PhD Intern	Cisco Systems, Inc.	May 2022 – Aug. 2022
Mentor: Directed Reading Program	UT Austin	Jan. 2018 – May 2022
Chair and Representative	Graduate Welfare Committee, UT Austin	Jun. 2020 – May 2022
Participant: Quantum Ideas Summer School	Duke University	Jun. 2019
Teaching Assistant: Quantum Computing	UT Austin	Jan. 2019 – Dec. 2019
Teaching Assistant: Physics Lab for Engineers	UT Austin	Sep. 2017 – Dec. 2018
Visiting Researcher	The Institute of High Energy Physics	Jul. 2016 – Aug. 2017

AWARDS AND FELLOWSHIPS

Vector Research Grant Vector Institute	Fall 2023 – Fall 2025
CQIQC Postdoctoral Fellowship UofT	Fall 2023 – Fall 2025
QuICS Hartree Postdoctoral Fellowship UMD (declined)	Spring 2023
Professional Development Award UT Austin	Spring 2023, Fall 2019
Lawrence C. Biedenharn Jr. Endowed Fellowship UT Austin	Fall 2017 – Spring 2018
Highest Honors at Graduation , UCSB	Summer 2016
Dean's Honors , UCSB	Fall 2013 – Spring 2016

PROFESSIONAL SERVICE

Below is a list of conferences and journals for which I have served as a reviewer, along with the initial invitation date:

Referee <i>Nature Communications</i>	Spring 2026
Referee <i>Quantum</i>	Fall 2025
Referee <i>npj Quantum Information</i>	Spring 2025
Referee <i>PRX Quantum</i>	Spring 2025
Referee <i>EPJ Quantum Technology</i>	Fall 2024
Referee <i>Physical Review A</i>	Fall 2024
Referee <i>Scientific Reports</i>	Summer 2024
Referee <i>Quantum Machine Intelligence</i>	Summer 2024
Referee <i>Theory of Quantum Computation, Communication, and Cryptography</i>	Spring 2024
Referee <i>Physical Review Research</i>	Fall 2023
Referee <i>Physical Review Letters</i>	Summer 2023
Referee <i>Physical Review B</i>	Spring 2023
Referee <i>Neural Networks</i>	Spring 2023
Referee <i>International Symposium on Symbolic and Algebraic Computation</i>	Fall 2023
Session Chair <i>APS March Meeting</i>	Spring 2023
Abstract Sorter <i>APS March Meeting</i>	Fall 2022
Referee <i>Quantum Information Processing</i>	Spring 2021

STUDENT MENTORSHIP

Shuyi Liang, PhD at NUS (starting 2026)	Research Paper, 2026
Brandon Barton, PhD at ETH Zürich	Research Paper, 2026
Jakub Garwola, PhD at University of Toronto	Research Paper, 2025
Xiang Zou, PhD at University of Toronto	Research Paper, 2025
Michelle Ding, undergraduate at UT Austin → PhD at UT Austin	Honors Thesis, 2025
Xuheng Zhao, master at ETH Zürich	Master Project, 2025
Ruize Ma, master at ETH Zürich → PhD at EPFL	Research Paper, 2025

Matteo D’Anna, PhD at ETH Zürich	Research Paper, 2025
Massimo Solinas, master at ETH Zürich → PhD at Regensburg University	Research Paper, 2025
Yifan Zhang, PhD at Princeton	Research Paper, 2024
Adam Martinez, undergraduate at UofT → PhD at Oxford	Honors Thesis, 2024
Xiaoxiao (Alice) Xiong, undergraduate at UBC → Caltech	Honors Thesis, 2023
Michelle Gelman, undergraduate at UT Austin → PhD at USC	Honors Thesis, 2023
Shahin Jahanbani, undergraduate at UT Austin → PhD at UCB	Research Paper, 2022

PRESENTATIONS AND INVITED TALKS

- “Verifiable Quantum Advantage: Why I’m Obsessed with Peaks,” ETH Zurich, Zurich, Spring 2026
- “Toward verifiable and practical quantum advantage,” NUS, Singapore, Spring 2026
- “Verifiable Quantum Advantage: Why I’m Obsessed with Peaks,” NTU, Singapore, Spring 2026
- “Verifiable Quantum Advantage: Why I’m Obsessed with Peaks,” Princeton University, Princeton, Spring 2026
- “A Physicist’s Guide to Quantum Complexity and Quantum Supremacy,” Spring School on Near-Term Quantum Computing, Benasque, Spring 2026
- “Verifiable quantum advantage with peaked circuit sampling,” Tsinghua University, Beijing, Winter 2025
- “Classical Simulability of Quantum Circuits with Shallow Magic Depth,” Year of Quantum Across Canada, Waterloo, Fall 2025
- “Co-design Research on Quantum Information Theory, Artificial Intelligence, and Quantum Chips,” HKUST(GZ), Guangzhou, Summer 2025
- “Artificial Intelligence & Quantum: empowering the future of computation”, Royal Bank of Canada, Halifax, Summer 2025
- “Compressing quantum dynamics with quantum machine learning,” Fields Institute, Toronto, Summer 2025
- “Observation of a non-Hermitian supersonic mode on a trapped ion quantum computer,” Tsinghua University, Beijing, Winter 2024
- “Observation of a non-Hermitian supersonic mode on a trapped ion quantum computer,” University of California, Berkeley, Online, Winter 2024
- “Variational quantum compilation for dynamical simulation,” University of California, Los Angeles, Online, Winter 2024
- “Variational quantum compilation for dynamical simulation,” Fudan University, Shanghai, Winter 2024
- “Variational quantum compilation for dynamical simulation,” Shanghai Jiao Tong University, Shanghai, Winter 2024
- “Observation of a non-Hermitian supersonic mode on a trapped ion quantum computer,” ETH Zurich, Zurich, Fall 2024
- “Scalable quantum dynamics compilation via quantum machine learning,” National University of Singapore, Online, Fall 2024
- “Scalable quantum dynamics compilation via quantum machine learning,” Vector Institute, Toronto, Fall 2024

- “Observation of a non-Hermitian supersonic mode on a trapped ion quantum computer,” University of Toronto, Toronto, Fall 2024
- “Advancing science in the era of quantum and AI”, IBM-CQIQC meeting, Toronto, Spring 2024
- “Qubit-efficient quantum simulation with sequential circuits,” Tsinghua University, Beijing, Winter 2023
- “Quantum volume for measurement-based quantum processors,” APS March Meeting, Las Vegas, Spring 2023
- “Quantum volume for photonic quantum computing,” Cisco Research, San Jose, Fall 2022
- “Holographic simulation of correlated electrons and thermal states on a trapped-ion quantum processor,” APS March Meeting, Chicago, Spring 2022
- “Holographic simulation of correlated electrons and thermal states on a trapped-ion quantum processor,” Brookhaven National Laboratory, Spring 2022
- “Interacting fermions on a quantum processor,” Quantum Circuits, Inc., New Haven, Spring 2022
- “QED driven QAOA for network-flow optimization,” Quantum Information Processing (QIP), Shenzhen, Winter 2020
- “Quantum computing today,” The Institute of High Energy Physics, Beijing, Summer 2019